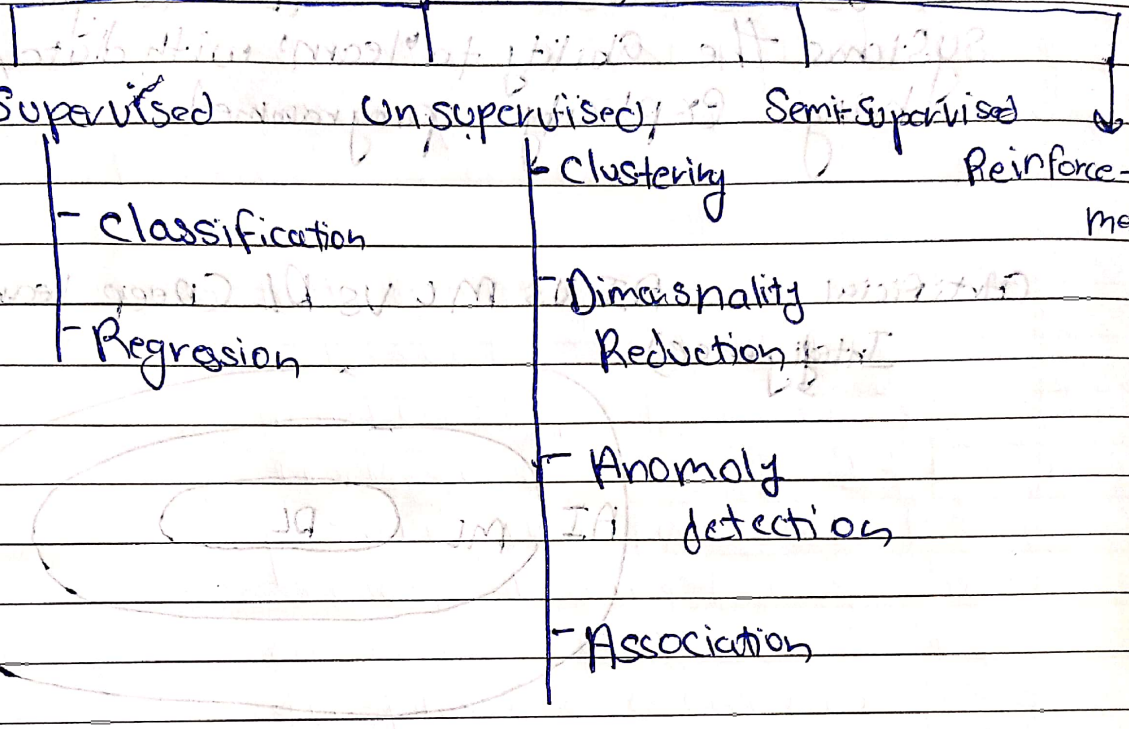
classmate
Date
Page

Types of ML



* Supervised learning : When we have both input or output and understand the relationship between both of them and then predict the output for new input.

Or

It is a type ML where a model learns from a labelled data, meaning each input has a correct output. The model compares its prediction with actual results and improves over time to increase accuracy

- Types of Supervised learning:

Regression: Where output is categorical/numerical values (age, weight, etc)

Classification - Where output is a categorial values (gender, national$, etc.)

* Unsupervised ML: opposite of supervised ML where we have only input not output. So in this we use

Clustering - We group the unlabelled data based on similarity and we label the group of category and process on it. Its goal is to discover pattern or relationships within the data without any prior knowledge of categories or label

Dimensity reduction: It is a process where we decrease no. of column/features or variables in a dataset while retaining as much of the original information as possible. It helps simplifying data, helps to analyze and visualize.

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anomly detection : Identifies unusual pattern in data, aiding aiding fraud detection, & cybersecurity and equipment failure.

Association Rule learning : It is used to discover interesting relation ships between variables in large datasets. It identifies pattern in a form of 'if-then' rules, Showing how the presence of some item in data implies the presence each other

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Unsupervised learning is a type of ML where the model works without labelled data. It learns pattern on its own by grouping similar data points or finding hidden structure without any human intervention.

classmate
Date
Page

Semi-supervised

* Semi-Supervised: It is a ML model that uses small amount of labelled data and large amount of unlabelled data to train models. Reduces labelling cost.

* Reinforcement learning: It is a type of ML where there is no input and output we start with scratch with help of interacting with enviornment with to maximise cumulative rewards.

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